NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 1 QUESTIONS AND SOLUTIONS.

SECTION A : 5 marks each

1. Last night was too cold, the temperature was -5^oC. It is now 6^oC. How many degrees warmer is it now?

Solution

We need to find the difference in temperature between now and last night.
$6-(-5)=6+5=11^\circ C$

2. If a machine produces 300 items in one minute, how many would it produce in 10 seconds?

Solution

10 seconds are equivalent to $\frac{10}{60}=\frac{1}{6}$ of a minute.
It implies that $\frac{1}{6}\times 300=50$ items are produced in 10 seconds.
Alternatively, If 300 items are produced in 60 seconds (a minute),
then $\frac{300}{60}=5$ items are produced in a second since we have 60 seconds in a minute.
It follows that $5\times 10=50$ items are produced in 10 seconds.

3. The mean of the numbers $4, 12, 9, 8, x, 2x$ is 8. What is the value of x?

Solution

$\frac{4+12+9+8+x+2x}{6}=8$. It follows that $33+3x=48$. Therefore $x=5$.

4. A new mathematical operation, $\oslash$, is defined by $a\oslash b=\dfrac{2a}{b}$. If $a\oslash 4=18$, what is the value of a?

Solution

$a\oslash 4=\frac{2a}{4}=18$. It follows that $2a=4\times 18$. Therefore $a=36$.

5. Maria's gadget takes five-digit positive integers as input. When the 5-digit integer $ABCDE$ is input, the gadget outputs the positive integer $A+B\times C+D\div E$. What will be the output when the input is 30263?

Solution

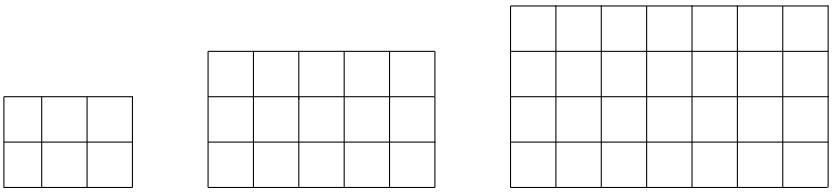
$3+0\times 2+(6\div 3)=3+0\times 2+2=3+(0\times 2)+2=3+0+2=5$

6. What fraction goes in the $\boxed{?}$ to make the equation $\dfrac{2}{3}+\dfrac{3}{4}+\dfrac{4}{5}=\dfrac{5}{6}+\dfrac{6}{7}+\boxed{?}$ true?

Solution

$\frac{2}{3}+\frac{3}{4}+\frac{4}{5}=\frac{40+45+48}{60}=\frac{133}{60}$ and $\frac{5}{6}+\frac{6}{7}=\frac{35+36}{42}=\frac{71}{42}$ It follows that
$\boxed{?}=\frac{133}{60}-\frac{71}{42}=\frac{1326}{2520}=\frac{221}{420}$.

7. Squares measuring 1cm by 1cm are arranged to form the following sequence of rectangles:



Rectangle 1 Rectangle 2 Rectangle 3

Many more rectangles are drawn, determine the perimeter of rectangle 5.

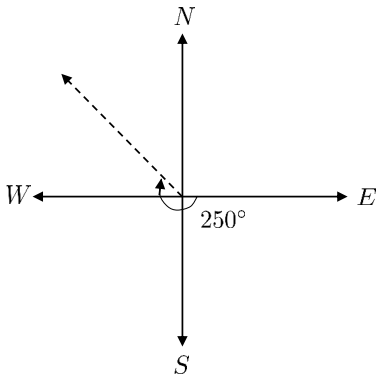
Solution

From the 3 rectangles drawn, each new rectangle has one more row and two more columns compared to the previous one. Rectangle 5 will have 6 rows and 11 columns.
The perimeter becomes $2(6 + 11) = 34cm$.

8. A bat is flying east. Sadly, it is knocked by an eagle which spins it 225^0 clockwise. What direction is the bat facing now?

Solution

Turning 225° from east will change the bat to be flying north-west as shown in Figure below;



9. My mathematics teacher master Lubega has 18 single socks. He puts on clean socks every three days. How many days can he wear these socks before he has to wash them all?

Solution

For 18 single socks, there are 9 pairs of socks and he wears each for 3 days, so he will need to wash them after $9 \times 3 = 27$ days.

10. The digits of the year 2024 total 8, that is, $2 + 0 + 2 + 4 = 8$. How many years will it be until this happens again?

Solution

To have $20xy$ digits sum to 8, $x + y = 8 - (2 + 0) = 6$. So the options for (x, y) are $(0, 6), (1, 5), (2, 4), (3, 3), (4, 2), ...$ The next year which gives that pattern is 2033 which corresponds to $(3, 3)$. This gives $2033 - 2024 = 9$ years.

SECTION B : (10 marks each)

11. At Monday's practice, Lydia sprints 200m 24 times. She rests for 30 seconds between each consecutive pair of sprints. Lydia sprints at a constant speed of 8m/s.

    (a) How many times does Lydia take the 30 seconds rest sessions?

    Solution

    Lydia rests for 30 seconds between the 1st and 2nd sprints, the 2nd and 3rd sprints, and so on up to the 23rd and 24th sprints. Thus Lydia takes the 30 seconds rest 23 times.

    (b) Determine Lydia's total practice time in minutes.

    Solution

    Since Lydia sprints at a constant speed of 8m/s, then it takes her $\frac{200m}{8} = 25$seconds to sprint 200m. She completes 24 such sprints and so her time which is spent on sprinting is $24 \times 25 = 600$seconds. Lydia also takes 23 rests, each of 30 seconds and so her time spent on resting is $23 \times 30 = 690$ seconds. Lydia total practice time is thus $600 + 690 = 1290$ seconds. This gives $\frac{1290}{60} = 21.5$ minutes.

12. Three tanks $T1, T2, T3$ contain water. The number of litres in each is shown in the table below.

$T1$	$T2$	$T3$
3500	1600	3900

    (a) Water is moved from tanks $T1$ and $T3$ into $T2$ so that each tank contains the same volume of water. How many litres of water are moved from $T1$ to $T2$?

    Solution

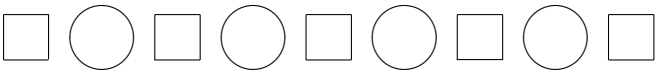
    For all to have the same volume, the volume must be equal to the average of the three tanks, thus $\frac{3500+1600+3900}{3} = \frac{9000}{3} = 3000$Litres in each. so move $3500 - 3000 = 500$ litres from $T1$ to $T2$.

    (b) From **12(a)** above, each tank now has the same volume of water and each tank is 0.25 full. How many litres must be added to each tank to make it full?

    Solution

    Let x be the total capacity of each tank, it follows that $\frac{1}{4}x = 3000$. It follows $x = 4 \times 3000 = 12000$. So add $12000 - 3000 = 9000$ Litres.

13. Place the integers $1, 2, 4, 5, 6, 9, 10, 11, 13$ in the circles and squares below with one number in each shape. Each integer must be used exactly once and the integer in each circle must be equal to the sum of the integers in the two neighbouring squares.



Solution

We could pick larger integers to go into the circles, i.e 13. Next to the filled circle of 13, we could fill the neighbouring squares 11 and 2. We could then choose 10 to go into the last circle with neighbouring squares filled as 1 and 9. Lastly fill integers $4, 5, 6$ appropriately with 6 being in the circle. The filled in shapes are depicted in Figure below;



Note: Alternatively, the shapes could be filled with the order given above reversed, i.e

$$9, 10, 1, 5, 4, 6, 2, 13, 11.$$

14. In a special sequence, the first term is a positive integer and each term after the first is determined in the following way:
- if a term, n, is odd, the next term is $n + 3$,
- if a term, n, is even, the next term is $n + 4$,
- if a term, n, is a prime, the next term is $n + 5$.

(a) If the first term in the special sequence is 51, what is the fifth term in the sequence?

Solution

Since 51 is odd, this gives a sequence of 51, 54, 58, 62, 66. The fifth term is 66.

(b) If the first term in the special sequence is 42, what is the fifth term in the sequence?

Solution

Since 42 is even, this gives a sequence of 42, 46, 50, 54, 58. The fifth term is 58.

(c) If the third term in the special sequence is 32, what are the possibilities for the first term?

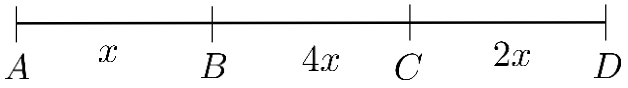
Solution

For the third term of 32, we have 3 candidates for second term, namely 29, 28 and 27. Since 29 is 3 steps before 32, it would be valid if it were an odd number yet not a prime. For 27, it is five steps before 32, it would be valid if it were a prime number. This leaves 28 as the only one which is valid since it is 4 steps before 32 and it is even, Whe the second term is 28, we have 3 options for the first term namely; 25, 24 and 23. Since 25 is 3 steps before 28 and it is an odd, it is valid. Also 24 is 4 steps before 28 and it is even, therefore valid. Lastly 23 is 5 steps before 28 and it is a prime, therefore valid. The possible sequences are 25, 28, 32, 24, 28, 32 and 23, 28, 32. The possibilities for the first term are 25, 24 and 23.

15. Points A, B, C, and D lie along a line, in that order. If $AB : AC = 1 : 5$, and $BC : CD = 2 : 1$. Find the ratio $AB : CD$.

Solution

Consider the Figure below



Let AB be x such that $x > 0$, Then $BC = 4x$ since $AB : AC = 1 : 5$, implying that $AC = 5x$ and $CD = 2x$ since $BC : CD = 2 : 1$. So $AB : CD = X : 2X = 1 : 2$.

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